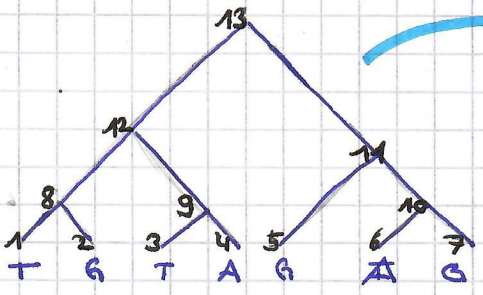
Fitch

1. Bottom-Up



Given tree

a) For each leaf ℓ, set $S(\ell) := \{s(c,\ell)\}$

$S(1) = \{s(A,1), s(G,1), s(T,1)\}$

$= \{0,0,1\} \hat{=} \{T\} = S(3) \cancel{= S(6)}$

$S(2) = \{G\} = S(5) = S(7)$

$S(4) = \{A\} = S(6)$

b) For each internal node, if the children share candidates, these shared candidates are candidates for the node. If the children do not share candidates, both sets of candidates (of the children) are candidates for the node.

$$S(8) = \begin{cases} S(1) \cap S(2) = \{\emptyset\} & \text{if } \{T\} \cap \{G\} \neq \emptyset \\ S(1) \cup S(2) = \{T,G\} & \text{otherwise} \end{cases} = \{T,G\}$$

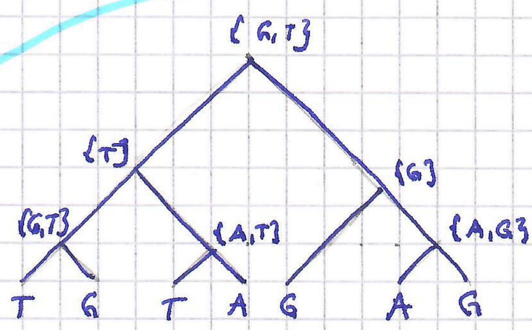
$S(9) = S(3) \cup S(4) = \{T,A\}$

$S(10) = S(6) \cup S(7) = \{A,G\}$

$S(11) = S(5) \cap S(10) = \{G\}$

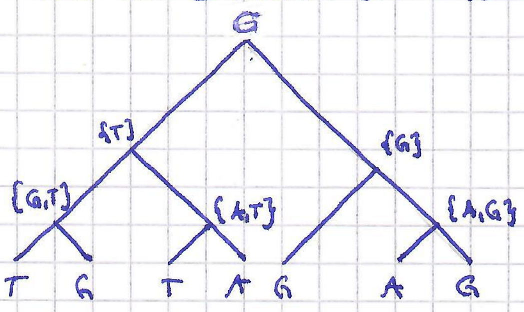
$S(12) = S(8) \cap S(9) = \{T\}$

$S(13) = S(11) \cup S(12) = \{G,T\}$



2. Top-Down Refinement

a) If the candidate set for the root contains more than one element, arbitrarily set the character state to one of them



b) (i) If the state assigned to an internal node is in ~~only~~ one of the candidate sets of it's children, set the character state of that child to the state of the internal node.
(ii) Otherwise, arbitrarily set the character state of the child to a state from its candidate set.